

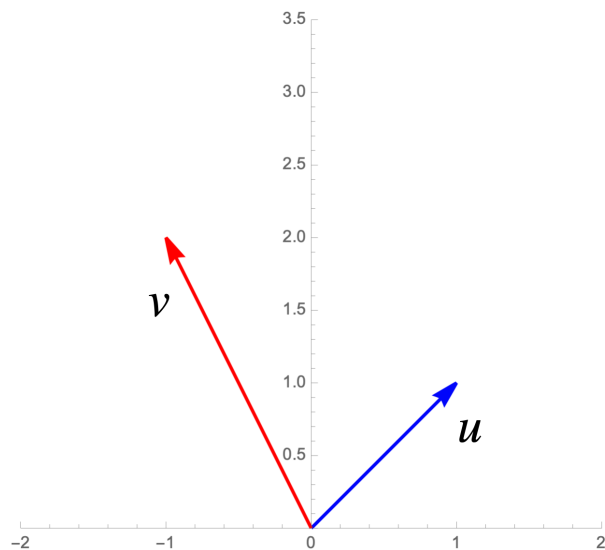
Warm-up 1

Using the standard basis, the vector v is given as $\begin{pmatrix} 5 \\ 1 \end{pmatrix}$. Write v using the basis

$b_1 = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, $b_2 = \begin{pmatrix} -7 \\ 5 \end{pmatrix}$, where these new basis vectors are also given in the standard basis.

During class 1

a) Using the ordered basis $u, v \in \mathbb{R}^2$ pictured below, write down the matrix representing the linear transformation that scales u by 2 and v by 3.



b) Now write down the matrix representing the same linear transformation, but using the standard basis for $\mathbb{R}^2$.